

SULABH KATILA

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EDUCATION

St. Joseph's University

B.S. in Mathematics & Computer Science, Minor in Data Science

May 2026

Valedictorian | GPA: 3.95/4.00

TECHNICAL SKILLS

Languages: C#, Python, Java, JavaScript, TypeScript, Bash, SQL, HTML5

Frameworks: .NET (dotnet), Playwright, Selenium, Flask, FastAPI, Django, React, Express, Next.js

AI/ML: Fine Tuning, RAG, MCP, LangChain, LangGraph, Vector-Database, Scikit-learn, PyTorch, TensorFlow, NumPy

Tools: Unix/Linux, Git, Docker, Kubernetes, Google Cloud Platform (GCP), NodeJS, MySQL, MongoDB, NoSQL

EXPERIENCE

TransPerfect

New York, NY

QA Automation Engineering Intern

February 2025 – May 2026

- Developed and maintained automated test framework using C# and .NET to support UI and API tests creation across projects
- Designed and developed custom code-analysis tool in Python to automatically update 400+ test cases, saving 80+ engineering hours
- Built NLP evaluation pipelines in Python/PyTorch to benchmark LLMs across specific tasks, directing production model selection
- Built and evaluated Retrieval-Augmented Generation (RAG) pipelines using vector retrieval and LLM-based reasoning to improve the accuracy and contextual relevance of AI-assisted engineering workflows
- Developed Model Context Protocol (MCP) integrations that enabled coding agents to securely retrieve documentation, query internal resources, and execute structured tool calls as part of an agentic development workflow
- Refactored legacy automated test suites plagued by flaky failures, cutting the false-positive rate by 65%, reducing CI/CD pipeline execution time by over 20 minutes per run, and lowering overall infrastructure resource usage

St. Joseph's University

New York, NY

Tutor

September 2023 – May 2026

- Tutored college Mathematics and Computer Science concepts to students, achieving up to an 80% improvement in test scores

Undergraduate Research Fellow

June 2024 – August 2024

- Conducted intensive exploratory data analysis using Python, Pandas, NumPy, and Matplotlib, interpreting trends and visualizations to identify borough-level market dynamics and translate statistical patterns into actionable research insights
- Built and compared machine learning models in Scikit-learn and TensorFlow, critically evaluating performance differences to determine which architectures best captured the nonlinear forces driving NYC real estate fluctuations
- Delivered an end-to-end independent research pipeline and presented findings to an academic audience at a research symposium

HealthStream

Remote, USA

QA Engineering Intern

September 2024 – December 2024

- Designed and built a Python/Selenium automated testing framework from scratch for a large-scale web application that had no existing automation, providing the team a scalable foundation to write and maintain tests going forward
- Performed manual QA across 20 feature tickets per sprint, filing up to 15 bug reports per sprint that surfaced critical defects

PROJECTS

BatMapper iOS:

- Built a camera-free iOS app in Swift and SwiftUI that maps indoor room geometry and floor plan in real time using echolocation
- Implemented a full acoustic signal processing pipeline (IIR bandpass filtering, cross-correlation, Gaussian smoothing, peak detection) to calculate wall distances from raw audio at 48kHz
- Fused acoustic distance data with IMU sensors (accelerometer + gyroscope) to construct a 2D floor plan as the user walks, including door detection and loop closure for drift correction

Smartphone Placement Detection:

- Engineered a real-time smartphone placement classification system that determines body-location (pockets, handheld, bag) using only raw accelerometer and gyroscope IMU data, achieving 88.5% accuracy
- Built a 156-feature DSP pipeline over 10-second IMU windows (FFT, autocorrelation, wavelets, Teager Energy, multiscale entropy), then applied MRMR feature ranking with wrapper-based selection to reduce dimensionality by 68% with no accuracy loss
- Built a Swift iOS app streaming live IMU data to a Python FastAPI backend running on-the-fly feature extraction to feed to a ported MATLAB-trained AdaBoostM2 model, with predictions broadcast via WebSocket

HONORS & AWARDS

- [Winner, HackNYU 2025](#)
- [Hon. Mention, CalHacks 12.0](#)
- 8 x Dean's List, SJNY
- Kappa Gamma Pi
- Delta Epsilon Sigma
- Esse Non Videri Scholarship, SJNY